REDBIRD FLIGHT SIMULATIONS
QUALIFICATION AND APPROVAL GUIDE (QAG)

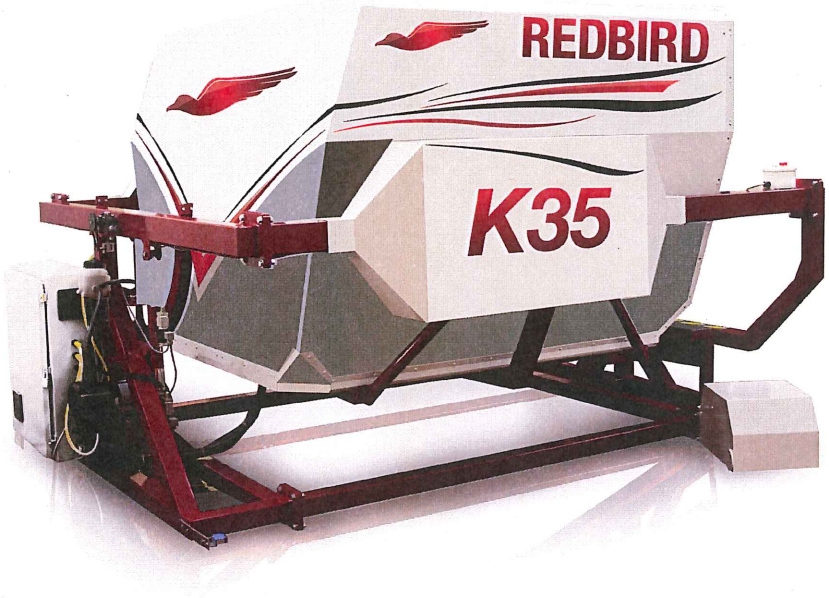
K35 RECERTIFICATION QUALIFICATION AND APPROVAL GUIDE, VERSION 1.0

FAA APPROVED QAG
Date 7/16/15

Signature John F Linsenmeyer

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Advanced Aviation Training Device

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SECTION 1: OVERVIEW STATEMENT

This Qualification and Approval Guide (QAG) includes a detailed description of all components, functions, and capabilities for the Redbird model K35/K9X. This includes any optional aircraft configurations with quality color pictures and diagrams. The required information as described in the current advisory circular, AC 61-136A / FAA Approval of Aviation Training Devices (ATD) and Their Use for Training and Experience, required for FAA approval, is contained within this document. This includes listing all the required qualifying items, functions, and capabilities. This QAG is used to clearly describe and verify the functionality of this aviation training device platform confirming its suitability for airman training and experience. A valid FAA Letter of Authorization (LOA) specifying the ATD allowances must also accompany the training device when utilized for airman training or experience requirements as specified in 14 CFR §61 or 141.

This ATD must continue to maintain its performance and function without degradation. The operator of this device is expected to maintain its condition and functionally when used for airmen training or experience requirements. Only the configurations approved for this model, as described within this QAG, can be utilized when satisfying FAA experience or training requirements. Any additions, changes, or revisions to the model or to the available configurations described within this document must be approved in writing by The General Aviation and Commercial Division, AFS-800.

Any questions concerning FAA approval of ATDs should be directed to the General Aviation and Commercial Division, AFS-800.

SECTION 2: OVERALL AVIATION TRAINING DEVICE (ATD) DESCRIPTION AND PICTURES

The Redbird model K35 is based on the dimensions and layout of a Hawker Beechcraft King Air production turboprop multi-engine land aircraft. This closely represents the overall functionality, performance, avionics, and instrumentation. The platform consists of a cockpit section, instructor's control station, visual display system and an audio system. It incorporates a combination of hardware and software components that is assembled and checked by Redbird Flight Simulations. All hardware elements are permanently installed and designed so the cockpit has the appearance and feel of an actual aircraft. From the pilot's seated position, there are no computer hardware elements such as keyboards, pointing devices, etc. for his or her use.

The K35 provides realistic and true-to-scale cockpit design, avionics, and reliable hardware/software performance. This platform also provides an effective training environment for student and certificated pilots. This includes the capability of practicing scenario based flight training events, simulated equipment failure and emergency procedures, pilot evaluations, instrument procedures/ experience, and facilitating increased pilot proficiency overall.

Multi Engine-Land, Turbo Prop Land with Enclosed Cockpit



Image 1: General view of Motion Platform and Cockpit Enclosure

The Redbird K35 and K9X are versatile and affordable devices that are designed to represent a variety of Beechcraft King Air Models. It is equipped with the following notable features:

- 3-axis electric motion platform providing pitch, roll and yaw motions.
- Wrap-around exterior visuals provided by 6 LCD screens
- Dual pilot controls including a 2-axis control-loaded yoke and interconnected rudder pedals
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to match the aircraft wherever possible
- An interchangeable instrument panel to allow development of future configurations
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructors station, allowing the instructor to operate from inside or outside the simulator
- Optional supplemental oxygen system for the pilot and/or co-pilot



Image 2: Interior view of Flight Deck Instruments



Image 3: General view of Pilot Yoke



Image 4: Detailed view of Pilot Yoke Switches (Left Side)



Image 5: Detailed view of Pilot Yoke Switches (Left Top)



Image 6: Detailed view of Pilot Yoke Switches (Center)

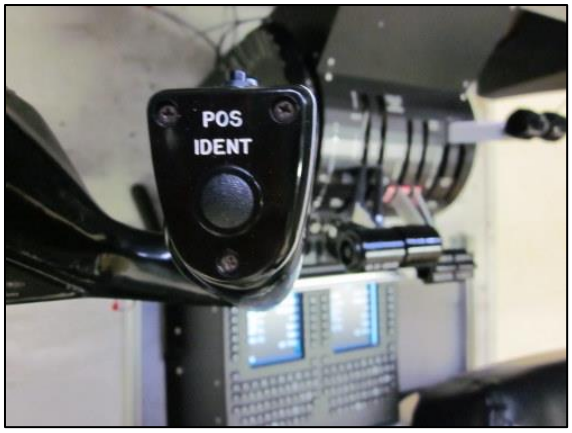


Image 7: Detailed view of Pilot Yoke Switches (Right Side)



Image 8: Detailed view of Co-Pilot Yoke



Image 9: Detailed view of Co-Pilot Yoke Switches (Left Side)



Image 10: Detailed view of Co-Pilot Yoke Switches (Center)



Image 11: Detailed view of Co-Pilot Yoke Switches (Right Side)

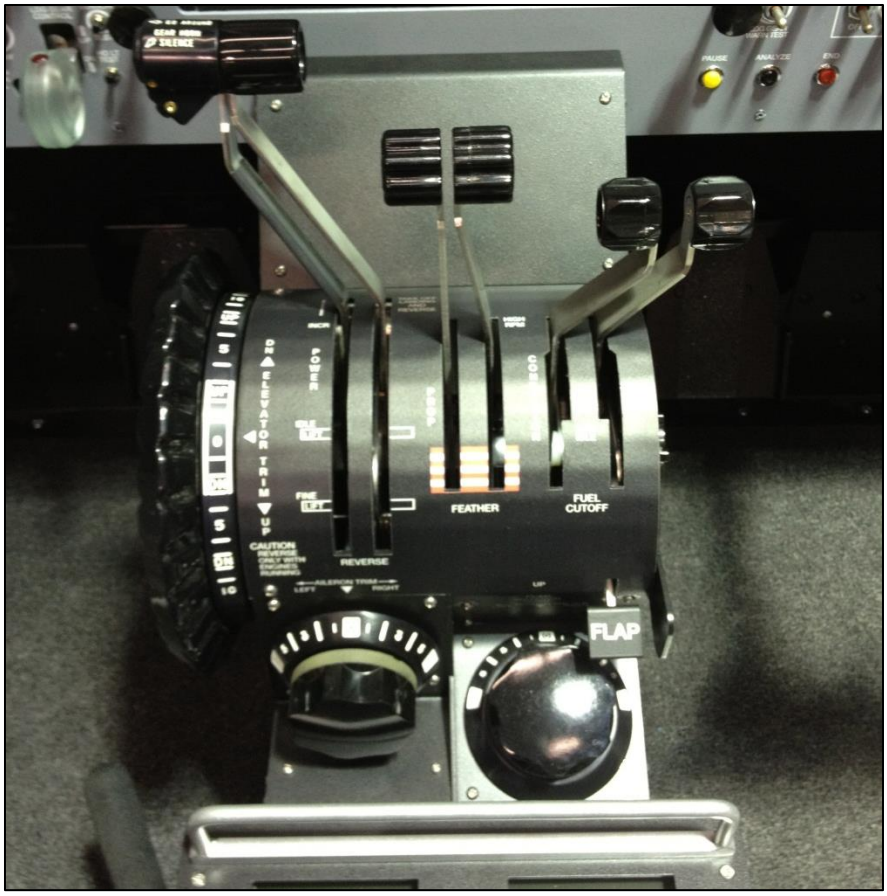


Image 12: Detailed view of Throttle Quadrant

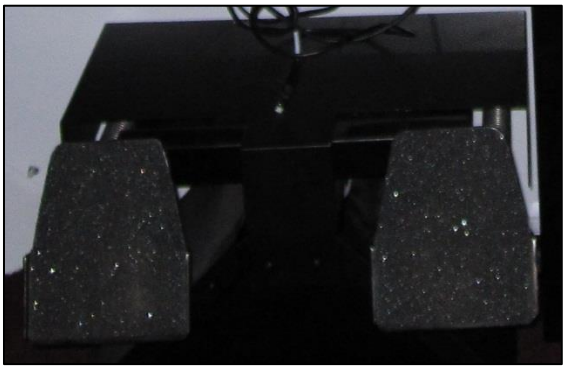


Image 13: Detailed view of Pilot/Co-Pilot Rudder Pedals



Image 14: Detailed of view of Dual FMS and Airplane Pressurization Controls



Image 15: General view of Fuel System Control Pilot Side Circuit Breaker Panel

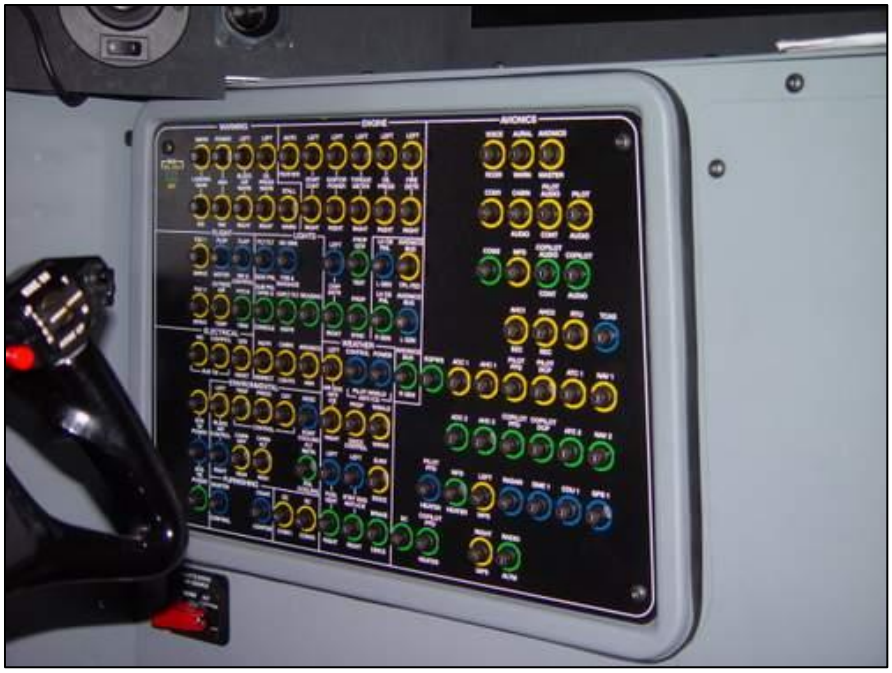


Image 16: General view of Co-Pilot Circuit Breaker Panel



Image 17: General view of Overhead Switch Panel



Image 18: General view of Pilot Side Lower Switch Panel (Right)

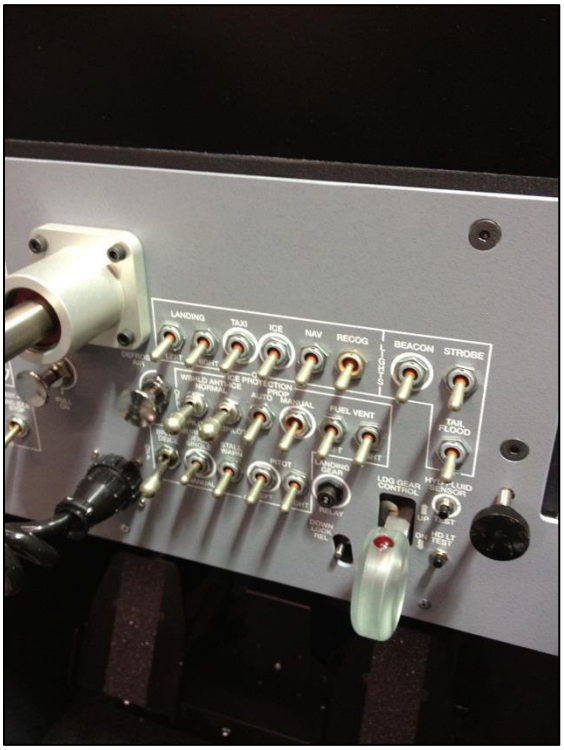


Image 19: General view of Pilot Side Lower Switch Panel (Right)



Image 20: General view of Co-Pilot Lower Switch Panel Co-Pilots (Left)



Image 21: Detailed view of Landing Gear Handle



Image 22: General view of Emergency Landing Gear Extension Handle



Image 23: General view of Instructors Station Tablet PC



Image 24: General interior view of Assembled Cockpit (Pilot Side)



Image 25: General interior view of Assembled Cockpit (Co-Pilot Side)

SECTION 3: TRAINING DEVICE COMPONENTS LIST

All Configurations Include:

Item	Component Name	Manufacturer	Type	Qty	Description/Function	Configuration
1	Motion Platform	Redbird Flight Simulations	Hardware	1	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	All
2	Cockpit Enclosure	Redbird Flight Simulations	Hardware	1	Cockpit enclosure to exclude distractions.	All
3	Dual Yoke	Redbird Flight Simulations	Hardware	2	Dual, Control loaded pitch and roll controller with switches and buttons for airplane systems operation.	All
4	Dual Rudder Pedals	Redbird Flight Simulations	Hardware	2	Dual Rudder control pedals with toe brakes.	All
5	Throttle Quadrant	Redbird Flight Simulations	Hardware	2	Twin, turbine engine controls with throttle lever, propeller pitch lever, condition lever, flap position lever and pitch, aileron and rudder trim controls with indicators for each.	All
6	Instrument Controls	Redbird Flight Simulations	Hardware	1	Flight instruments, radios, airplane configuration and systems controls.	All
7	Redbird Proline 21	Mindstar Aviation	Software	1	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	All
8	Redbird FMS 3000	Mindstar Aviation	Software	1	Virtual FMS display and logic.	All
9	Redbird SFDS	Mindstar Aviation	Software	1	Secondary Flight Display System. Displays backup flight instruments.	All
10	FMS Panel	Redbird Flight Simulations	Hardware	1	FMS controls, keypad and display screen. Includes airplane cabin pressurization system controls.	All
11	Switch Panel 1	Redbird Flight Simulations	Hardware	1	Lower switch panel with airplane configuration and systems controls.	All
12	Switch Panel 2	Redbird Flight Simulations	Hardware	1	Overhead switch panel with airplane systems controls.	All
13	Fuel System Control	Redbird Flight Simulations	Hardware	1	Pilot side panel with controls for airplane fuel system.	All

Item	Component Name	Manufacturer	Type	Qty	Description/Function	Configuration
14	Circuit Breaker Panel	Redbird Flight Simulations	Hardware	1	Pilot side circuit breaker panel.	All
15	Circuit Breaker Panel	Redbird Flight Simulations	Hardware	1	Co-pilot side circuit breaker panel.	All
16	Simulation Computer	Redbird Flight Simulations	Hardware	1	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
17	Windows 7	Microsoft	Software	1	Operating system.	All
18	ESP	Microsoft	Software	1	Simulation engine. (depending on serial number)	All
19	Prepar3D	Lockheed Martin	Software	1	Simulation engine. (depending on serial number)	All
20	RB Sim	Redbird Flight Simulations	Software	1	Simulation control and component integration.	All
21	Motion Control Computer	Redbird Flight Simulations	Hardware	1	Host computer for FMX Control.	All
22	FMX Control	Redbird Flight Simulations	Software	1	Motion system control and component integration. (depending on serial number)	All
23	LCD	22” Industry Standard	Hardware	6	Flat Panel displays for exterior views.	All
24	LCD	19” Industry Standard	Hardware	3	Flat Panel displays for virtual instruments.	All
25	Instructors Station Computer	Industry Standard	Hardware	1	Host computer for Instructors Station.	Optional
26	Instructors Station	Redbird Flight Simulations	Software	1	Environmental, location and failure controls with map, track and glideslope display.	All
27	Parrot	Redbird Flight Simulations	Software	1	Optional software package that links iOS aviation or navigational apps to Redbird Devices	Optional
28	Cygnus	Redbird Flight Simulations	Software	1	Optional software package providing ATC Services	Optional
29	Insight	Redbird Flight Simulations	Software	1	Optional package that provides instructors live video feed of the cockpit and instrumentation through Prepar3D software	Optional

Item	Component Name	Manufacturer	Type	Qty	Description/Function	Configuration
30	Instrument Procedures Database	RealNav Data	Software	1	Provides for FAA published instrument navigation procedures, database per 14 CFR 97 (enroute, approach)	All

Table 1: Training Device Component List

STATEMENTS OF COMPATIBILITY OF SOFTWARE AND HARDWARE AC61-136A, APPENDIX 2, 4(B):

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows 7 (Operating System)

"This is to certify that Microsoft Corporation, the owner and developer of the Windows 7 operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Microsoft ESP (Simulation Engine)

"This is to certify that Microsoft Corporation, the owner and developer of the ESP Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Lockheed Martin Prepar3D (Simulation Engine)

This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

SECTION 4: AVIATION TRAINING DEVICE (ATD) DESIGN CRITERIA LIST

The following section provides the detailed “word for word” listing and design criteria of each of the required items, functions, and capabilities (listed in AC 61-136A, for BATD requirements Appendix 2 and the additional AATD items of Appendix 3) and operational performance value/scale (as applicable) for each of the functions described for the Redbird K35/K9X.

DESCRIPTION OF COMPLIANCE ITEMS AND FUNCTIONS

BATD DESIGN CRITERIA

All configurations for this model, as noted, meet AC 61-136A, Appendix 2 requirements

a. Redbird K35 model meets the following Control Input requirements.

(1) Physical flight and aircraft control systems are designed such that they are recognizable as to their function and how they are manipulated solely from their appearance. No keyboard, mouse, or gaming joystick is used to control the aircraft.

(2) Virtual controls are limited to setting the aircraft configuration, location, weather conditions, and pausing action.

(3) No keyboard or mouse is used to set or position any of the below listed features of this device. The required and additional equipment is operated in the same manner in which it would be operated in the aircraft represented. **This includes power levers, propeller levers, condition levers, landing gear, and wing flaps.**

(4) The physical arrangement, appearance, and operation of controls, instruments, and switches closely represents the model of aircraft including: **master switch, ignition and start switches, battery, generators, EMER BATT switch, external power switch, avionics master, beacon/strobe lights, navigation lights, taxi and landing lights.** Also included are: **pitot heat, anti-ice switches, bleed air switches, fire detection switches, autofeather test, overspeed test, standby fuel pumps, fuel crossflow, and AUX transfer override.**

(5) Only the software package evaluated and approved by the FAA is available for use on this computer system.

b. “Redbird K35” model meets the following Control Requirements:

(1) Airplane

(a) **Self-centering control yoke or control stick** that allows continuous adjustment of pitch and bank

(b) **Self-centering rudder pedals** rudder pedals that allow continuous adjustment of yaw and corresponding reaction in heading and roll.

(c) **Power Levers** that allows for continuous movement from idle to full power, with corresponding changes in pitch and yaw.

(d) **Condition, propeller, and throttle/power control(s)** as applicable to the model aircraft represented

(e) **Controls** for the following items applicable to the aircraft represented:

- Wing flaps,
- Pitch trim,
- Communication and navigation radios,
- Clock or timer,
- Gear handle (if applicable),
- Transponder,
- Altimeter,

c. Redbird K35/K9X model meets the following Control Input requirements:

(1) Time from control input to recognizable system response must be without delay (i.e., not appear to lag in any way). Redbird Flight Simulations Incorporated verifies that the K35 meets this requirement.

(2) The control inputs are tested by the computer and software at each session startup and displayed as a confirmation message or a “warning message” that the transport delay time or any design parameter is out of original tolerances. It is not possible to continue the training session unless the problem is resolved and all components are functioning properly. This test considers the items listed in the display requirements.

d. Redbird K35/K9X model meets the following Display Requirements:

(1) **Instruments and indicators** replicated and properly located as appropriate to the aircraft model represented:

Flight instruments in a standard configuration, represented as traditional “round dial” flight instruments, or as an electronic primary flight instrument display (PFD) and multi-function display (MFD) with reversionary and back-up flight instruments.

- A sensitive **altimeter** with incremental markings each 20 feet or less, operable throughout the normal operating range of the aircraft or family of aircraft represented.
- A **magnetic direction indicator**
- A **heading indicator** with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the aircraft or family of aircraft represented.
- An **airspeed indicator** with incremental markings as shown on the aircraft or family of aircraft represented; airspeed markings of less than 40 knots need not be displayed.
- A **vertical speed indicato**r with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ± 2,000 fpm total display, or as applicable to the aircraft or family of aircraft being represented.
- A **gyroscopic rate-of-turn indicator or equivalent** with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.
- A **slip and skid indicator** with coordination information displayed in the conventional skid ball format where a coordinated flight condition is indicated with the ball in the center position. **A split image triangle indication** may be used if applicable to the aircraft or family of aircraft being represented.
- An **attitude indicator** with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to the aircraft or family of aircraft represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.
- **Engine instruments** as applicable to the aircraft or family of aircraft being represented, providing markings for normal ranges and minimum and maximum limits.
- A **pneumatic pressure gauge** or instrument pressure gauge, as applicable, with a display applicable to the aircraft or family of aircraft represented.
- A **flap setting indicator** that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.
- A **pitch trim indicator** with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.
- **Communication radio(s)** with display(s) of the radio frequency in use.
- A **clock** with incremental markings for each minute and second, or a timer with a display of minutes and seconds,
- A **transponder** that displays the current transponder setting.
- A **fuel quantity indicator(s)** that displays the fuel remaining, either in analog or digital format,

as appropriate for the aircraft or family of aircraft represented.

Navigation Equipment: Replicates both precision and non-precision instrument navigation, including approach procedures (with an aural identification feature if normally available), and a marker beacon receiver. Examples: an ILS receiver, non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or Very high frequency Omnidirectional Range (VOR). Graduated markings as indicated below are present on each course deviation indicator as applicable.

Markings include:

- One-half dot or less for **course/glide slope deviation indicator** (i.e., VOR, LOC, or ILS),
- 5 degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable,

(2) All instrument displays listed above must be visible during all flight operations. The update rate of all displays must provide an image of the instrument that:

(a) Does not appear to be out of focus or illegible
(b) Does not appear to “jump” or “step” to a distracting degree during operation.
(c) Does not appear with distracting jagged lines or edges.
(d) Does not appear to lag relative to the action and use of the flight controls.

(3) Control inputs are reflected by the flight instruments in real time and without a perceived delay in action. Display updates or actions show all changes (within the total range of the replicated instrument) that are equal to or greater than the following:

(a) Airspeed indicator: change of 5 knots.
(b) Attitude indicator: change of 2 degrees in pitch and bank.
(c) Altimeter: change of 10 feet.
(d) Turn and bank: change of ¼ standard rate turn.
(e) Heading indicator: change of 2 degrees.
(f) Vertical speed indicator (VSI): change of 100 fpm.
(g) Tachometer: change of 25 rpm or 2 percent of turbine speed.
(h) VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS.
(i) ADF: change of 2 degrees.
(j) GPS: change as appropriate for the model of GPS based navigator represented.
(k) Clock or timer: change of 1 second.

(4) **Flight Displays reflect proper dynamic behavior**. Examples: VSI reading of 500 fpm reflects a corresponding movement in altimeter. Throttle movement reflects an appropriate change in pitch, rpm indication, or the power indicator. Pitch changes reflect an appropriate change in airspeed and altitude.

e. Redbird K35/K9X model meets the following Flight Dynamics requirements:

(1) All flight dynamics are comparable to the way the represented airplane performs and handles.

(2) All simulated aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate and hovering/sideward/forward/rearward flight) are comparable to the airplane represented. A performance table is included in the QAG for each aircraft configuration for sea level and 5,000 feet.

(3) Vertical lift component changes as a function of bank/load factor, comparable to the aircraft represented.

(4) Changes in flap setting and gear position are accompanied by changes in flight dynamics comparable and reasonably equivalent to the aircraft represented.

(5) The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft or family of aircraft represented performs and handles.

f. Redbird K35/K9X model meets the following Instructional Management Requirements:

(1) The instructor can pause the system at any point during the training session for the purpose of administering instruction regarding the training task at hand.

(2) If a training session begins with the “aircraft in the air” and ready for the performance of a particular procedural task, the instructor can manipulate the following system parameters independently of the simulation:

- Aircraft geographic location,
- Aircraft heading,
- Aircraft airspeed,
- Aircraft altitude, and
- Wind direction, speed, and turbulence.

(3) The **system is capable of recording** both a horizontal and vertical track of aircraft movement for the entire training session for later playback and review.

(4) The instructor can disable any of the instruments prior to or during a training session and simulate failure of any of the instruments without stopping the flight simulation to affect the failure.

(5) The ATD has an available **REAL NAV Approach Database** that is local to the training facility. All navigational data must be based on **procedures as published per 14 CFR part 97.**

AATD DESIGN CRITERIA

All configurations, as noted in AC 61-136A, Appendix 3 meet these *additional* AATD design criteria items listed.

a. Redbird K35/K9X model meets the following additional AATD CRITERIA.

(1) A realistic shrouded (enclosed) or unshrouded (open) cockpit design and instrument panel arrangement representing a specific model aircraft cockpit.

(2) **Cockpit knobs, system controls, switches, and/or switch panels in realistic sizes and design** appropriate to each intended functions, in the proper position and distance from the pilot's seated position, and representative of the category and class of aircraft being represented.

(3) Primary flight and navigation instruments appropriately sized and properly arranged that exhibit neither stepping nor excessive transport delay.

(4) Digital Avionics Panel

(5) IFR Moving Map **GPS Navigation System**

(6) **Two-axis autopilot**, and, as appropriate, a flight director (FD). (Airplane only requirement)

(7) **Pitch trim** (manual or electric pitch trim) permitting indicator movement either electrically or analog in an acceptable trim ratio.

(8) **Visual system** providing cues for both day and night, including variable VFR and IFR conditions including:

- Adjustable visibility parameters; and
- Adjustable ceiling parameters.

(9) **Cockpit seat** is representative of the aircraft make/model.

(10) A **separate instructor station** that permits effective interaction without interrupting the flight in overseeing the pilot's horizontal and vertical flight profiles in real time and space. This includes the ability to:

(a) Oversee tracks along published airways, holding entries and patterns, and Localizer (LOC) and glideslope (GS) alignment/deviation (or other approaches with a horizontal and vertical track).
(b) Function as air traffic control in providing vectors, etc., change in weather conditions, ceilings, visibilities, wind speed and direction, light/moderate/severe turbulence, and icing conditions.

(c) Invoke failures in navigation and instruments, radio receivers, landing gear and flaps, engine power (partial and total), and other aircraft systems (pitot, electric, static, etc.) by using either a keyboard or mouse.

(12) Push to talk switch on the control yoke or stick

(13) Capable of simulating all of the emergency procedures for which a checklist is prescribed in the aircraft manufacturers' pilot operating handbook (POH) or flight manual

SECTION 5: AIRCRAFT CONFIGURATIONS

PREVIOUSLY APPROVED CONFIGURATIONS:

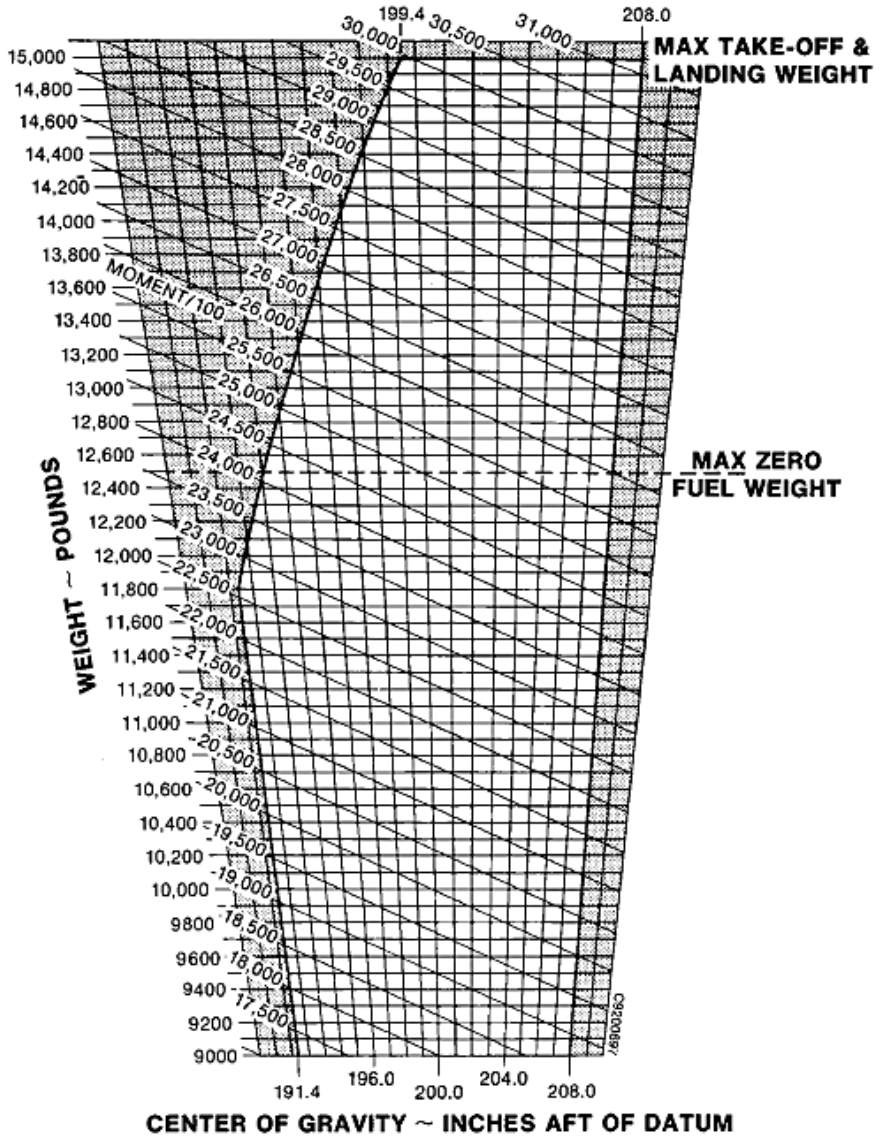
1. **Redbird K35-C1 (Hawker Beechcraft King Air 350)**
   a. **Cockpit**



I**mage 26:** K35 Instrument Panel Overview

b. Weight & Balance Data

MOMENT LIMITS VS WEIGHT



MOMENT LIMITS VS. WEIGHT

WEIGHT (LB)	MINIMUM MOMENT/ 100 (LB-IN)	MAXIMUM MOMENT/ 100 (LB-IN)	WEIGHT (KG)	MINIMUM MOMENT/ 100 (KG-IN)	MAXIMUM MOMENT/ 100 (KG-IN)
9000	17226	18270	4000	7656	8320
9100	17417	18928	4050	7752	8424
9200	17609	19136	4100	7847	8528
9300	17800	19344	4150	7943	8632
9400	17992	19552	4200	8039	8736
9500	18183	19760	4250	8135	8840
9600	18374	19968	4300	8230	8944
9700	18566	20176	4350	8326	9048
9800	18757	20384	4400	8422	9152
9900	18949	20952	4450	8517	9256
10000	19140	20800	4500	8613	9360
10100	19331	21008	4550	8709	9464
10200	19523	21216	4600	8804	9568
10300	19714	21424	4650	8900	9672
10400	19906	21632	4700	8996	9776
10500	20097	21840	4750	9092	9880
10600	20288	22048	4800	9187	9984
10700	20480	22256	4850	9283	10088
10800	20671	22464	4900	9379	10192
10900	20863	22672	4950	9474	10296
11000	21054	22880	5000	9570	10400
11100	21245	23088	5050	9666	10504
11200	21437	23296	5100	9761	10608
11300	21628	23504	5150	9857	10712
11400	21820	23712	5200	9953	10816
11500	22011	23920	5250	10049	10920
11600	22202	24128	5300	10144	11024
11700	22394	24336	5350	10240	11128
11800	22585	24544	5352	20244	11132
11900	22806	24752	5400	10350	11232
12000	23028	24960	5450	10461	11336

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MOMENT LIMITS VS. WEIGHT

WEIGHT (LB)	MINIMUM MOMENT/ 100 (LB-IN)	MAXIMUM MOMENT/ 100 (LB-IN)	WEIGHT (KG)	MINIMUM MOMENT/ 100 (KG-IN)	MAXIMUM MOMENT/ 100 (KG-IN)
12100	23250	25168	5500	10572	11440
12200	23473	25376	5550	10683	11544
12300	23696	25584	5600	10795	11648
12400	23920	25792	5650	10907	11752
12500	24144	26000	5700	11019	11856
12600	24368	26208	5750	11132	11960
12700	24594	26416	5800	11244	12064
12800	24819	26624	5850	11357	12168
12900	25045	26832	5900	11471	12272
13000	25272	27040	5950	11584	12376
13100	25499	27248	6000	11698	12480
13200	25727	27456	6050	11812	12584
13300	25955	27644	6100	11927	12688
13400	26184	27872	6150	12041	12792
13500	26413	28080	6200	12156	12896
13600	26642	28288	6250	12272	13000
13700	26873	28496	6300	12387	13104
13800	27103	28704	6350	12503	13208
13900	27334	28912	6400	12619	13312
14000	27566	29120	6450	12735	13416
14100	27798	29328	6500	12852	13520
14200	28031	29536	6550	12969	13624
14300	28264	29744	6600	13086	13728
14400	28498	29952	6650	13204	13832
14500	28732	30160	6700	13321	13936
14600	28966	30368	6750	13439	14040
14700	29202	30576	6800	13558	14144
14800	29437	30784	6804	13567	14152
14900	29673	30992			
15000	29910	31200			

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c. Performance Data

Altitude	MTOW	Max Operating Speed (V_{MO})	Max Cruise Speed (V_C)	Air Minimum Control Speed (V_{MCA})	Stall Speed (V_S)	Max Power Rate of Climb (Dual)	Max Power Rate of Climb (SE)
Sea Level	15,500 lbs.	263 KIAS	257 KTAS at 1500 RPM	94 KIAS with Flaps-Up	81 KIAS with Flaps-Down	2750 FPM at 100 KIAS	750 FPM at 125 KIAS
5,000 feet	15,500 lbs.	263 KIAS	270 KTAS at 1500 RPM	94 KIAS with Flaps-Up	81 KIAS with Flaps-Down	1400 FPM at 100 KIAS	540 FPM at 125 KIAS

Table 2: K35 Performance Data
(Note: Data represents Standard Atmospheric Conditions at maximum gross weight)

ADDITIONAL CONFIGURATIONS INCLUDED IN THIS VERSION (1.1):

1. **Redbird K9X-C1 (Hawker Beechcraft King Air 90)**
   a. Cockpit

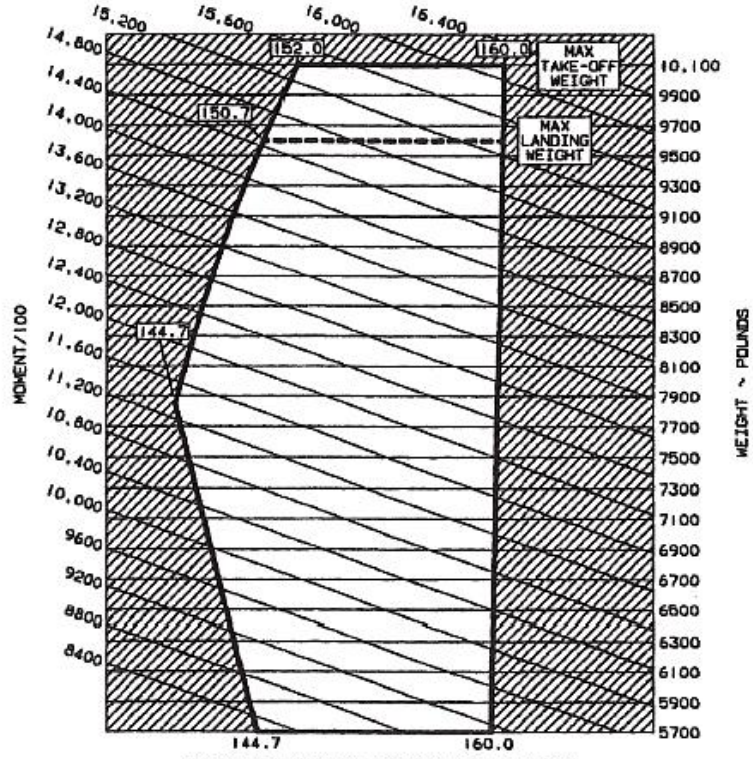


Image 27: K9X Instrument Panel Overview

b. Weight & Balance Data

Raytheon Aircraft Company
Model C90GT

MOMENT LIMITS VS. WEIGHT



ENVELOPE BASED ON THE FOLLOWING WEIGHT AND CENTER OF GRAVITY LIMIT DATA (LANDING GEAR DOWN)

WEIGHT CONDITION	FORWARD CG LIMIT	AFT CG LIMIT
10,100 POUNDS (MAXIMUM TAKE-OFF)	152.0	160.0
9600 POUNDS (MAXIMUM LANDING)	150.7	160.0
7850 POUNDS OR LESS	144.7	160.0

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USEFUL LOAD WEIGHT AND MOMENTS OCCUPANTS

WEIGHT (LBS)	PILOT OR COPILOT	CLUB SEATING CHAIR PASSENGER			AISLE FACING STORAGE SEAT	LAVATORY PASSENGER
	F.S. 129	F.S. 168	F.S. 215	F.S. 218	F.S. 243	F.S. 285
	MOMENT/100 (LB-IN)					
80	103	134	172	174	194	228
90	116	151	194	196	219	257
100	129	168	215	218	243	285
110	142	185	236	240	267	314
120	155	202	258	262	292	342
130	168	218	280	283	316	370
140	181	235	301	305	340	399
150	194	252	322	327	364	428
160	206	269	344	349	389	456
170	219	286	366	371	413	485
180	232	302	387	392	437	513
190	245	319	408	414	462	542
200	258	336	430	436	486	570
210	271	353	452	458	510	599
220	284	370	473	480	535	627
230	297	386	494	501	559	656
240	310	403	516	523	583	684

USEFUL LOAD WEIGHT AND MOMENTS

CABINET CONTENTS				BAGGAGE CONTENTS			
WEIGHT (LBS)	FWD CABIN	AFT CABIN		WEIGHT (LBS)	AFT COMPT F.S. 277	WEIGHT (LBS)	AFT COMPT F.S. 277
	F.S. 153	F.S. 226	F.S. 243		MOMENT/ 100 (LBS-IN)		MOMENT/ 100 (LBS-IN)
	MOMENT/100 (LBS-IN)						
5	8	11	12	10	28	190	526
10	15	23	24	20	55	200	554
15	23	34	36	30	83	210	582
20	31	45	49	40	111	220	609
25	38	56	61	50	139	230	637
30	46	68	73	60	166	240	665
35	54	79		70	194	250	693
40		90		80	222	260	720
				90	249	270	748
				100	277	280	776
				110	305	290	803
				120	332	300	831
				130	360	310	859
				140	388	320	886
				150	416	330	914
				160	443	340	942
				170	471	350	970
				180	499		

NOTE

Floor loading shall not exceed 100 pounds per square foot.

CABIN CARGO CONVERSION

The cabin chairs are removable for cargo conversion. Cargo tiedown rings may be installed in the seat tracks. Floor loadings of 200 pounds per square foot are permissible on pallets installed on the seat tracks. It is the responsibility of the operator to ensure that the airplane is properly loaded.

c. Performance Data

Altitude	MTOW	Max Operating Speed (V_{MO})	Max Cruise Speed (V_C)	Air Minimum Control Speed (V_{MCA})	Stall Speed (V_S)	Max Power Rate of Climb (Dual)	Max Power Rate of Climb (SE)
Sea Level	10,100 lbs.	226 KIAS	225 KTAS at 1900 RPM	85 KIAS with Flaps-Up	78 KIAS with Flaps-Down	1990 FPM at 112 KIAS	490 FPM at 108 KIAS
5,000 feet	10,100 lbs.	226 KIAS	239 KTAS at 1900 RPM	85 KIAS with Flaps-Up	78 KIAS with Flaps-Down	1880 FPM at 112 KIAS	400 FPM at 108 KIAS

Table 3: K9X Performance Data
(Note: Data represents Standard Atmospheric Conditions at maximum gross weight)

SECTION 6: VISUAL SYSTEM WITH VFR, IFR, DAY AND NIGHT CAPABILITIES

Mindstar Aviation Redbird Proline 21 (Virtual Instrumentation)
"This is to certify that Mindstar Aviation, the owner and developer of the Redbird Proline 21, and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Redbird Proline 21 software package is fully compatible with the Redbird Flight Simulations K35/K9X and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Mindstar Aviation Redbird FMS 3000 (Virtual FMS)
"This is to certify that Mindstar Aviation, the owner and developer of the Redbird FMS 3000, and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Redbird FMS 3000 software package is fully compatible with the Redbird Flight Simulations K35/K9X and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Mindstar Aviation Redbird SFDS (Secondary Flight Display System)
"This is to certify that Mindstar Aviation, the owner and developer of the Redbird SFDS, and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Redbird SFDS software package is fully compatible with the Redbird Flight Simulations K35/K9X and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Redbird K35/K9X Visual System
The K35/K9X visual system is capable of providing a field-of-view of a minimum of 45 degrees horizontally and 30 degrees vertically, simultaneously for each pilot, including adjustable cloud base and visibility in night, dusk and day scenes.

- The Redbird K35/K9X provides a means of recording the visual response time for the visual system that is installed.
- The Redbird K35/K9X visual system is free of optical discontinuities and artifacts that create non-realistic cues.
- The visual system is directly displayed on six (6) LCD monitors inside the cockpit enclosure, situated in an arc around the Pilot. Each monitor is 28 cm tall, and 47 cm wide. Based upon the designated Pilot Eye Point, these monitors provide a horizontal FOV of 220 degrees and a vertical FOV of 30 degrees.

Daylight: The visual system provides full color presentations and sufficient surfaces with appropriate textural cues to conduct a visual approach, landing and airport movement. Surface shading effects are consistent with the simulated sun position.

Twilight: The visual system provides full color presentations of reduced ambient intensity, sufficient surfaces with appropriate textural cues that include self-illuminated objects such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement. Scenes include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by representative ownship lighting.

Night: The visual system provides the same as above except the portrayal of reduced ambient intensity, therefore, there is no ground cues that are not self-illuminating or illuminated by ownship lights.

Designated Eye Point: The designated Pilot Eye Point is located 52 cm from the center of the forward most external view monitor, 61 cm from the left most external view monitor and 24 cm from the ceiling of the simulator enclosure. This point is roughly centered over the pilot's seat when it is adjusted to the forward most position, at a height consistent with the height of the pilot's head.

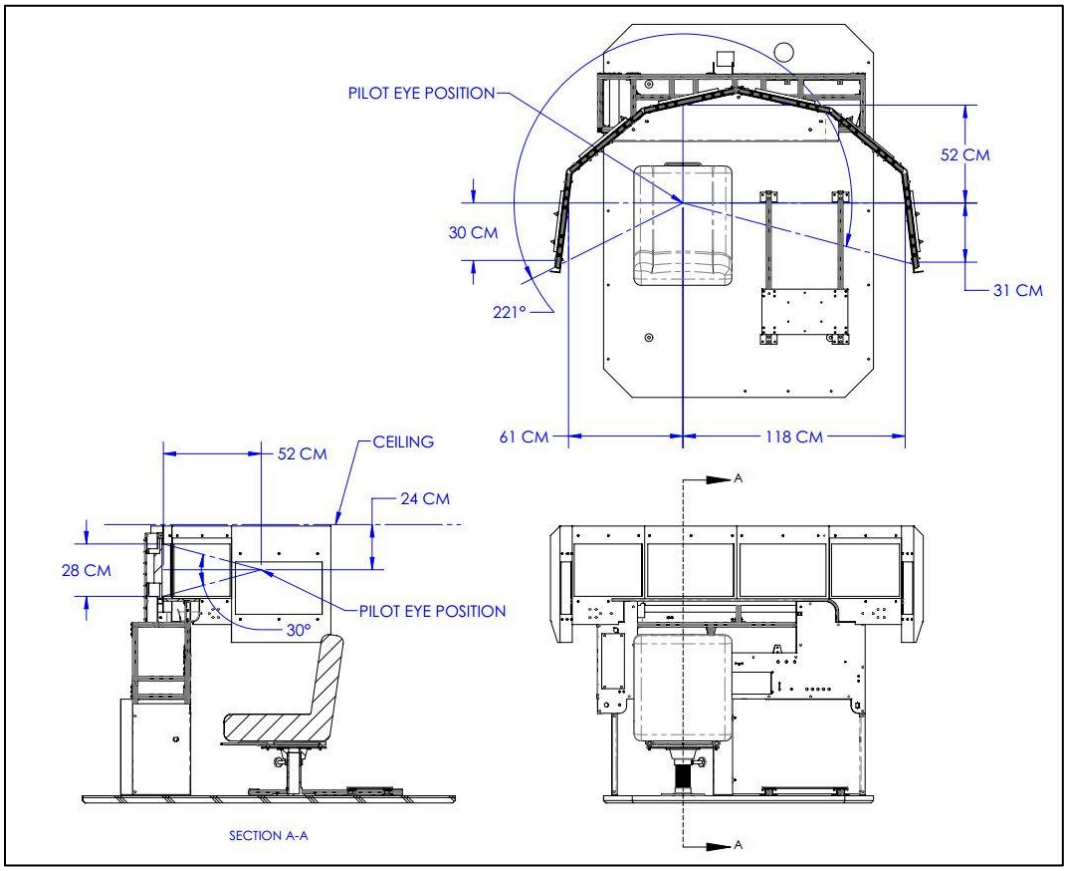


Figure 1: Designated Eye Point Diagram

SECTION 7: ATD FUNCTIONS AND MANEUVERS CHECKLIST

ATD FUNCTIONS AND MANEUVERS CHECKLIST

AVIATION TRAINING DEVICE ACTION CHECKLIST

Functions and Maneuvers	Yes, No, N/A
a. Preparation for Flight	
(1) Flight Deck Preflight	Yes
b. Pre-Takeoff	
(1) Engine start	Yes
(2) Brake operation	Yes
(3) Taxi operations and markings	Yes
c. Takeoff	
(1) **AIRPLANE Takeoff**	
(i) Power plant run-up and checks	Yes
(ii) Acceleration characteristics	Yes
(iii) Nose wheel and rudder steering	Yes
(iv) Effect of headwind/crosswind	Yes
(v) Instrument check	Yes
(vi) Landing gear, wing flap operation	Yes
(2) **HELICOPTER only Takeoff**	
(i) Power plant checks	N/A
(ii) From Hover	N/A
(iii) From Ground	N/A
(iv) Vertical	N/A
(v) Running	N/A
d. In-Flight Operation	
(1) **AIRPLANE In-Flight Operation**	
Runway and Takeoff	
(a) Normal and Maximum Performance Takeoffs	Yes
(b) Short and Soft Field Takeoff and Departures	Yes
(c) Normal and Maximum Performance Climbs	Yes
(d) Engine Failure and Emergency Procedures	Yes
Cruise	
(a) Performance Characteristics (speed vs. power)	Yes
(b) Normal, Climbing, and Descending Turns	Yes
(c) Performance Steep Turns	Yes

(d) Approach to Stalls, i.e. Stall Warning, Cruise, Takeoff & Approach and Landing Configurations	Yes
(e) In Flight Engine Shutdown (Multi-Engine only)	Yes
(f) In Flight Engine Start (Multi-Engine only)	Yes
Approach and Landing	
(a) Normal and Partial/No Flap Approach and Landings	Yes
(b) Short and Soft Field Approach and Landings	Yes
(c) Single Engine Approach and Landing (Multi-Engine)	Yes
Functions and Maneuvers	**Yes, No, N/A**
(2) **HELICOPTER In Flight Operations**	
(i) Hovering	N/A
(a) Forward	N/A
(b) Rearward	N/A
(c) Sideward	N/A
(d) Turns	N/A
(ii) Climb	N/A
(iii) Cruise	N/A
(a) Performance Characteristics	N/A
(b) Turns	N/A
(i) Recovery	N/A
(ii) Skidding	N/A
(iii) Slipping	N/A
(c) In Flight Engine Shutdown (Multi-Engine)	N/A
(d) In Flight Engine Start	N/A
(iv) Descent	N/A
e. Instrument Approaches	
(1) **Non-Precision**	
(i) GPS- WAAS (optional)	Yes
(ii) GPS- LAAS (optional)	Yes
(iii) All Engines Operating	Yes
(iv) One or More Engines Inoperative (as applicable)	Yes
(v) Approach Procedures (NDB, VOR, DME Arc, LOC/BC, LOC, LDA, SDF, ASR, LNAV/VNAV, GPS and LPV)	Yes
(2) **Precision**	
(i) ILS	Yes
(ii) Effects of Crosswind	Yes
(iii) With engine(s) inoperative	Yes
(iv) Missed Approach	Yes
(A) Normal	Yes
(B) With Engine(s) inoperative (as applicable)	Yes
(C) From Steep Glide Slope	Yes
f. Surface Operations (AIRPLANE-Post Landing)	

(1) Landing roll	Yes
(2) Braking Operation	Yes
(3) Reverse thrust Operation, if applicable	Yes
g. Any Flight Phase	
(1) **Aircraft and Power Plant Systems Functions and Simulated Failures**	
(i) Electrical	Yes
(ii) Flaps (Airplane)	Yes
(iii) Fuel & Oil	Yes
(iv) Landing Gear	Yes
(2) **Flight Management and Guidance Systems**	
(i) Two Axis Auto Pilot (AATD only)	Yes
Functions and Maneuvers	**Yes, No, or N/A**
(ii) Flight Director (AATD only) / System Displays	Yes
(iii) Navigation Systems and optional display configurations	Yes
(iv) Stall Warning Avoidance (Airplane)	Yes
(v) Multi-Function Displays PFD/MFD	Yes
(3) **Airborne Procedures**	
(i) Holding	Yes
(4) **Simulated Turbulence in Flight**	
(5) **Engine Shutdown and Parking**	
(i) Systems operation	Yes
(ii) Parking Brake Operation (Airplane)	Yes
h. Training device capable of replicating any emergency procedures provided in the Aircraft Flight Manual "checklists" for the model or aircraft configuration represented.	Yes

Any changes or modifications to this device which have not been reviewed, evaluated, and approved in writing by General Aviation and Commercial Division, AFS-800 may terminate FAA approval of this aviation training device.